

CSA1017 – SOFTWARE ENGINEERING

INDIVIDUAL REFLECTION REPORT

INTELLIGENT DISASTER EARLY WARNING AND EVACUATION MANAGEMENT SYSTEM

NAME: BALAJI K

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ROLE: Emergency Notification and Communication System

DESIGN / DEVELOPMENT DECISION:

I contributed to the emergency notification and communication component of the system. My focus was on designing how disaster warnings and emergency instructions can be delivered to citizens and authorities. I considered alert information such as disaster type, severity, affected location, recommended actions, and issuing authority.

CHALLENGE FACED AND SOLUTION:

The main challenge was ensuring that emergency messages are clear and reach users quickly. I addressed this by organising notifications according to severity and providing concise instructions. I also considered different communication methods so that users can receive important warnings even when one communication channel is unavailable.

LEARNING GAINED:

I learned the importance of real-time communication in disaster-management software. I understood that emergency messages must be accurate, simple, timely, and easy to understand. I also learned how notification services can interact with other modules of a software system.

SDG RELEVANCE:

The project supports SDG 11 by improving access to emergency information and helping communities respond to disaster situations. Effective communication can support safer decision-making during emergencies.

COURSE OUTCOME CONTRIBUTION:

My contribution supports course outcomes related to system communication, software integration, user requirements, interface design, and real-time information delivery. I learned how different system components can work together to deliver emergency alerts.

POSSIBLE IMPROVEMENT:

I would improve the notification system by adding SMS, voice calls, multilingual alerts, mobile push notifications, emergency broadcasts, acknowledgement tracking, and offline communication alternatives.